

COMP 3700A: Contributing to Open Source

~~Exploring OSS projects~~

Git Concepts

Charlotte Curtis
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Today's Goals

- Understand a bit more about Git
- Look at the projects you shared for homework:
 - Common documents
 - Project activity
 - Communication channels

References:

- "Open Source: What It Is and How to Contribute" — [Chapter 4 \(Git\)](#)

Housekeeping

- Lab grades will be updated on D2L out of seven
- Lab deadlines now made explicit on [labs README](#), with extensions for first two
- [Course notes](#) are now live
- [Project part 1](#) is now published

I asked Gemini and it was completely willing to invent answers. Please don't waste my time (and yours) outsourcing this to an LLM.

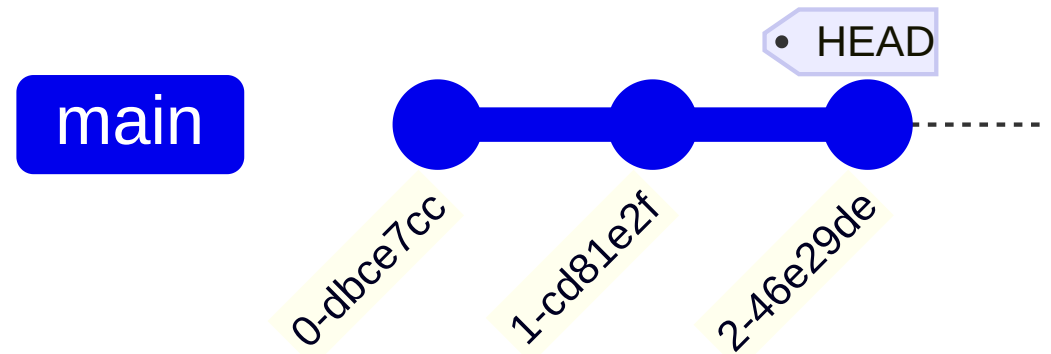
Single user, single branch

- `init` or `clone` to get repo
- For each commit:

```
git add whatever changed
git commit -m "Short message \
describing change"
```

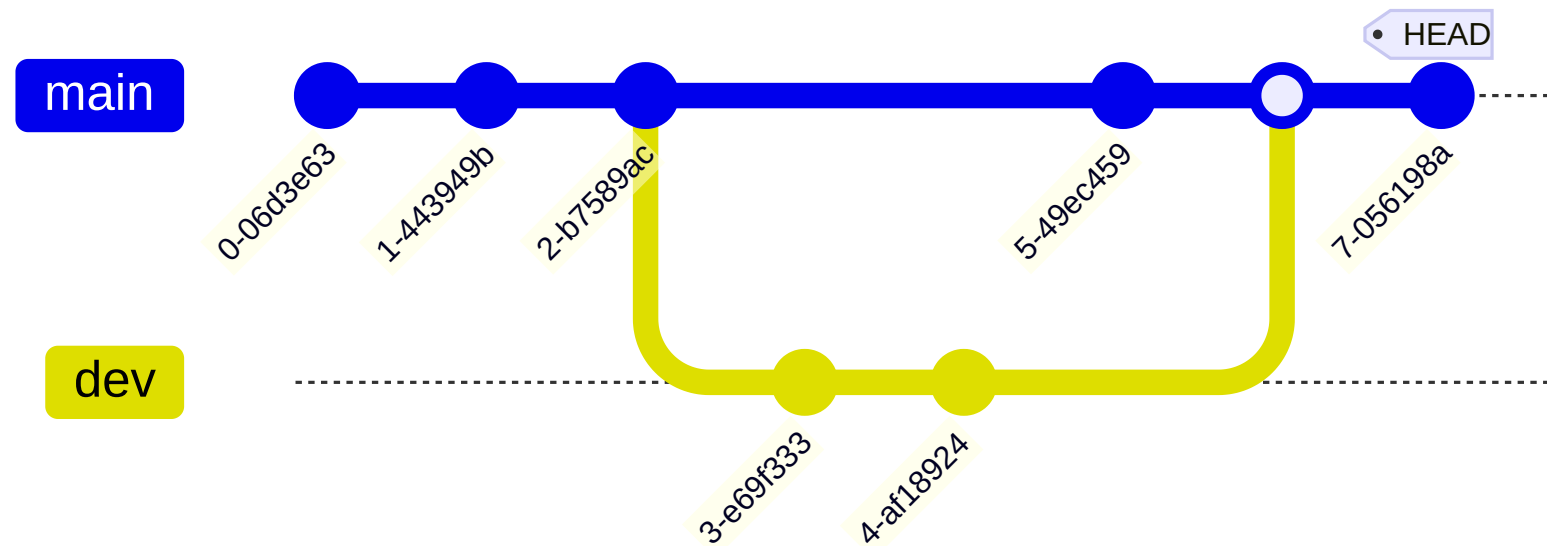
- Other useful commands:

```
git status
git log
git reset
```

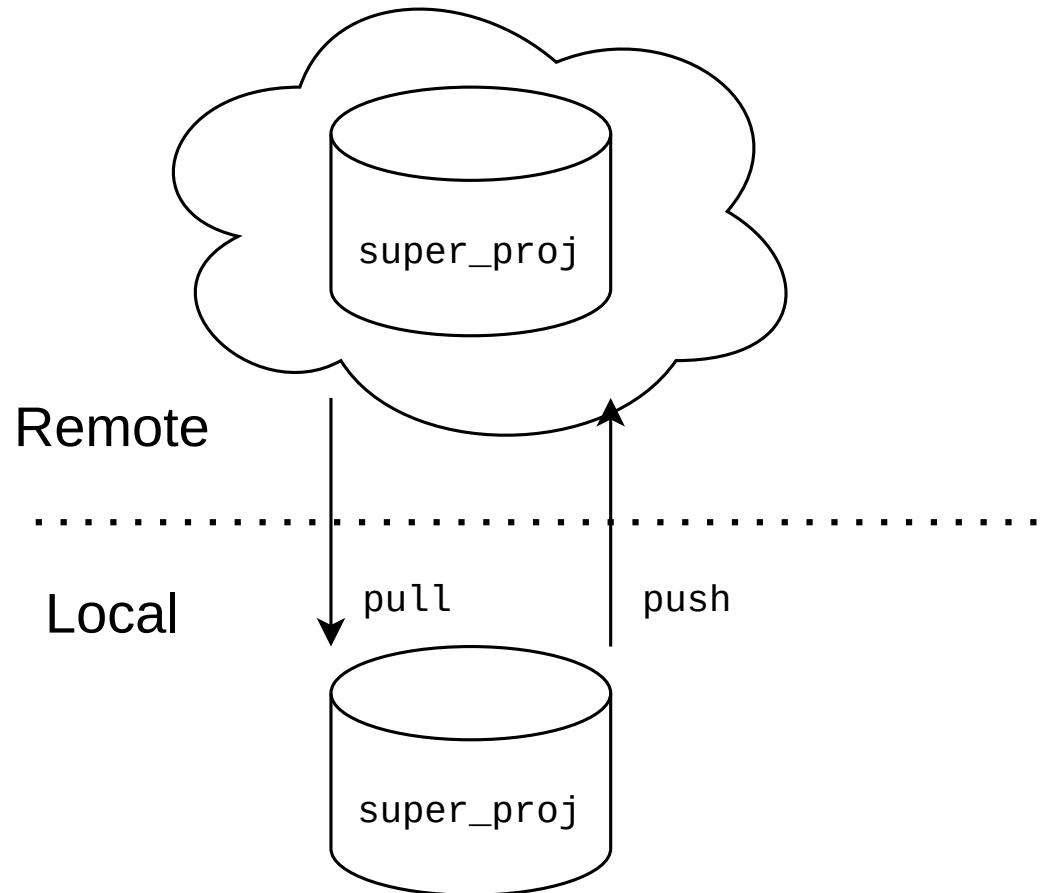


Single user, multiple branches

- `git checkout -b dev` (shortcut for `git branch dev && git checkout dev`)
- `git checkout main`
- `git merge dev`

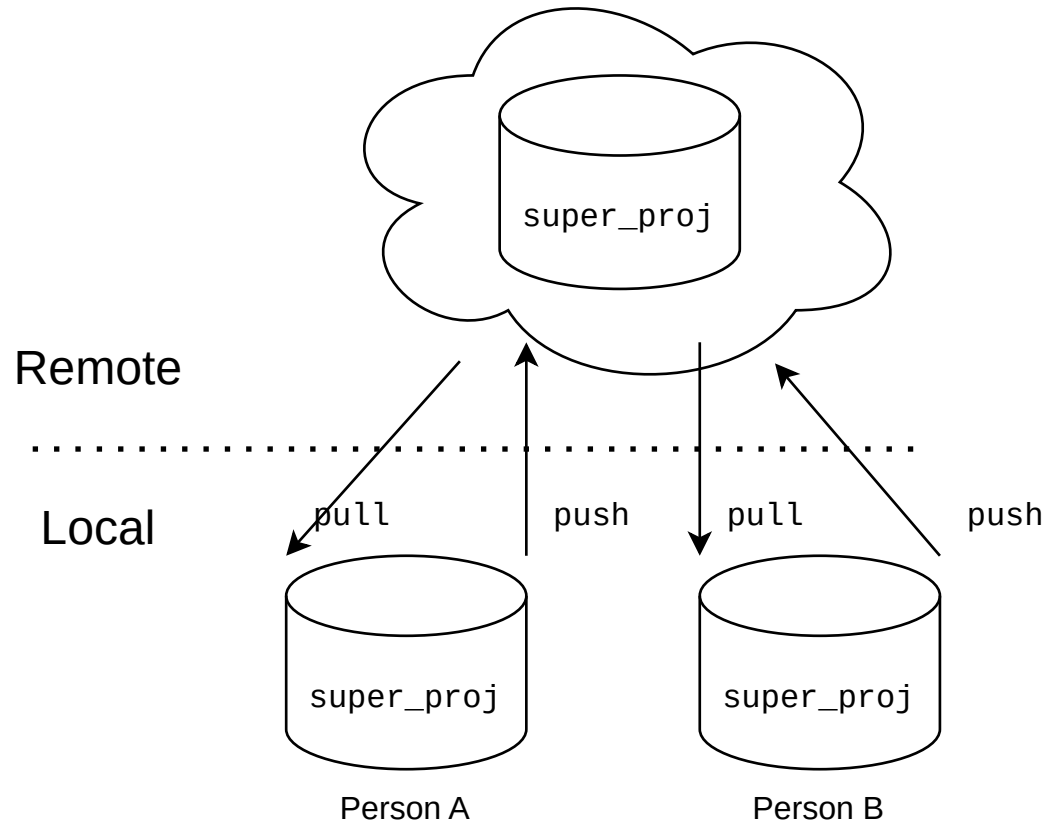


Adding a remote



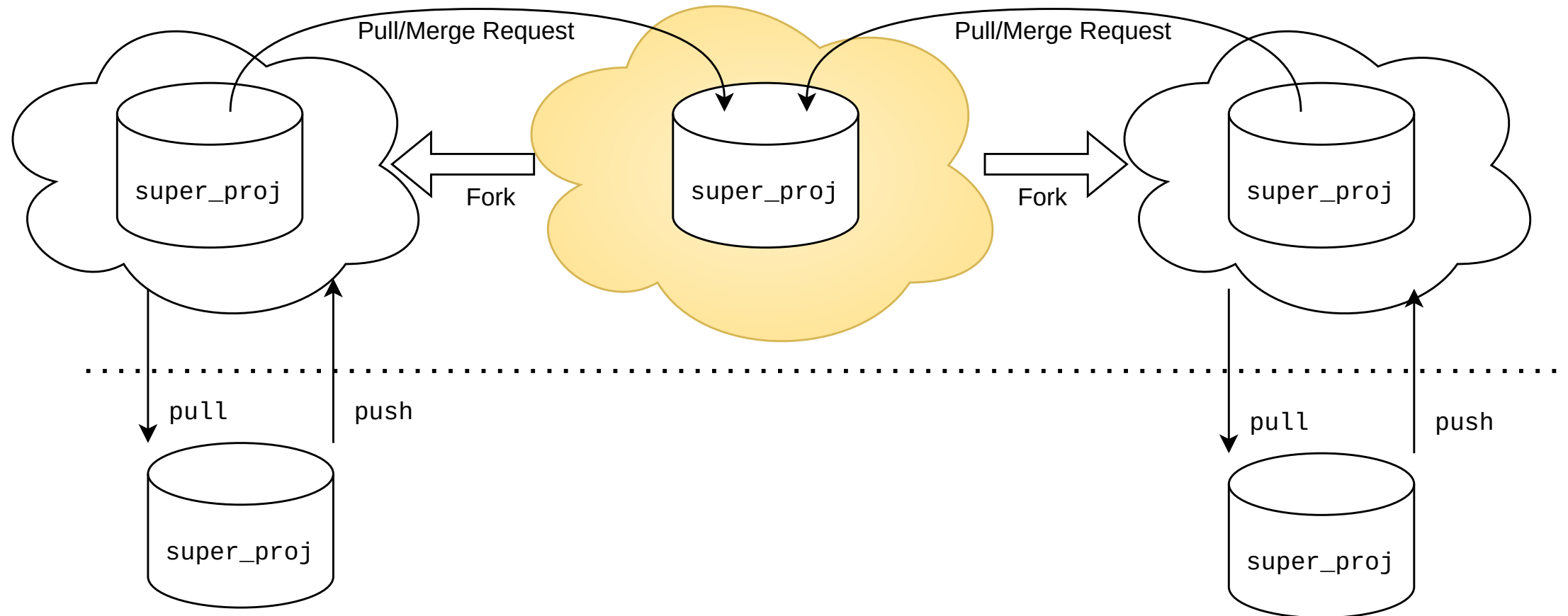
- `git remote add <remote_name> <url>`
- If you cloned from URL at the start, the remote `origin` should exist
- `git remote -v` shows all defined remotes and their roles

Multiple users, single remote, single branch



- Same commands, but now risk of **merge conflicts**
- Might need `git rebase`
- `git pull` is actually short for `git fetch` then `git merge` or `git rebase`, depending on configuration

Multiple users, multiple forks, single branch



Multiple users, multiple forks, multiple branches

- The most common setup in FOSS development
- One "canonical" repo, each developer has a fork
- Pull/merge requests (PRs) are **branch-specific**
 - Updates to a branch with an open PR will be added to the PR
 - Typical workflow: create a new branch for every new feature/bug fix

This is where I had you jump into the deep end last Thursday

Exploring communities — Wednesday

- With the projects you posted on the last [homework discussion](#):
 - What are some common documents?
 - How do you evaluate project activity?
 - How is communication handled?

Homework: Sign up to write [class notes](#) for a topic